

Custom Connectors

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Mastering Custom Connectors in Power Apps & Power Automate

Overview

This training curriculum is designed to help learners understand, build, and implement **Custom Connectors** in Microsoft Power Platform.

The course progresses from API fundamentals to advanced enterprise governance and deployment scenarios, with hands-on exercises for every concept.

Learning Objectives

By the end of this course, learners will be able to:

- Understand REST APIs and JSON structure
 - Create Custom Connectors from scratch
 - Configure authentication (API Key, Basic, OAuth 2.0)
 - Use Custom Connectors in Power Automate flows
 - Use Custom Connectors in Power Apps Canvas Apps
 - Implement error handling and versioning
 - Deploy connectors using solutions
 - Apply governance and DLP policies
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Module 1: API Foundations (Prerequisite Knowledge)

1.1 What is an API?

Description

An API (Application Programming Interface) allows two applications to communicate with each other. Power Platform uses APIs to connect to external systems.

Reference Links

- Web API Overview
<https://learn.microsoft.com/en-us/aspnet/web-api/>

- REST API Introduction
<https://restfulapi.net/>

Hands-On Exercise

Open in browser:

<https://jsonplaceholder.typicode.com/posts/1>

Observe the JSON response and identify:

- Endpoint
 - Data format
 - Key-value pairs
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1.2 REST API Architecture

Description

REST APIs use HTTP methods and resources to perform operations.

Reference Links

- REST Architecture Guide
<https://restfulapi.net/rest-architecture/>
- HTTP Status Codes
<https://developer.mozilla.org/en-US/docs/Web/HTTP/Status>

Hands-On Exercise

Using Postman:

- Send GET request to retrieve all posts
 - Send GET request for a single post
 - Compare responses
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1.3 HTTP Methods

Description

Method | Purpose -- | -- GET | Retrieve data POST | Create data PUT | Update data
DELETE | Delete data

Reference Links

- HTTP Methods Documentation
<https://developer.mozilla.org/en-US/docs/Web/HTTP/Methods>

Hands-On Exercise

Send a POST request with JSON body using Postman and observe the response.

1.4 JSON Structure

Description

APIs send and receive data in JSON format.

Reference Links

- JSON Guide
<https://www.json.org/json-en.html>
- JSON Formatter Tool
<https://jsonformatter.org/>

Hands-On Exercise

Copy API response and identify:

- Objects
 - Arrays
 - Nested properties
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1.5 API Authentication

Description

APIs use authentication mechanisms such as:

- API Key
- Basic Authentication
- OAuth 2.0

Reference Links

- API Security Best Practices
<https://learn.microsoft.com/en-us/azure/architecture/best-practices/api-design#authentication>
- OAuth 2.0 Overview
<https://oauth.net/2/>

Hands-On Exercise

Register on:

<https://openweathermap.org/api>

Generate an API key and test the API in browser or Postman.

Module 2: Introduction to Custom Connectors

2.1 What is a Custom Connector?

Description

A Custom Connector allows Power Platform to connect to any external REST API that does not have a built-in connector.

Reference Links

- Custom Connector Overview
<https://learn.microsoft.com/en-us/connectors/custom-connectors/>

Hands-On Exercise

Navigate to Power Apps → Custom Connectors

Explore existing connectors and review settings.

Module 3: Creating a Custom Connector

3.1 Create Custom Connector from Blank

Reference Links

- Create from Blank Guide
<https://learn.microsoft.com/en-us/connectors/custom-connectors/define-blank>

Hands-On Exercise

Create a connector:

- Name: WeatherConnector
- Configure Host

- Configure Base URL
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3.2 Define Actions and Parameters

Reference Links

- Define Actions
<https://learn.microsoft.com/en-us/connectors/custom-connectors/define-actions>
- Define OpenAPI Definition
<https://learn.microsoft.com/en-us/connectors/custom-connectors/define-openapi-definition>

Hands-On Exercise

Create Action:

- Name: GetWeather
- Method: GET
- Add Query Parameter: city
- Import sample response

Test the connector.

3.3 Testing Custom Connector

Reference Links

- Test Custom Connector
<https://learn.microsoft.com/en-us/connectors/custom-connectors/test-custom-connector>

Hands-On Exercise

Use the Test tab:

- Provide sample city
 - Execute request
 - Troubleshoot errors
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Using: Microsoft Power Automate

4.1 Create Flow Using Custom Connector

Reference Links

- Create Cloud Flow
<https://learn.microsoft.com/en-us/power-automate/get-started-logic-flow>
- Use Custom Connector in Flow
<https://learn.microsoft.com/en-us/connectors/custom-connectors/use-custom-connector>

Hands-On Exercise

Create Instant Flow:

- Add Custom Connector action
 - Store data in SharePoint
 - Send email notification
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4.2 Error Handling in Flow

Reference Links

- Error Handling in Power Automate
<https://learn.microsoft.com/en-us/power-automate/error-handling>

Hands-On Exercise

Add:

- Scope
 - Configure Run After
 - Condition for non-200 response
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Module 5: Using Custom Connector in Power Apps

Using: Microsoft Power Apps

5.1 Add Custom Connector to Canvas App

Reference Links

- Use Custom Connector in Canvas App
<https://learn.microsoft.com/en-us/power-apps/maker/canvas-apps/register-custom-api>

Hands-On Exercise

Create Canvas App:

- Add TextInput (City)
 - Add Button
 - Call API on OnSelect
 - Display result in Label or Gallery
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Module 6: Authentication Deep Dive

6.1 API Key Configuration

Reference:

<https://learn.microsoft.com/en-us/connectors/custom-connectors/define-security>

Hands-On:

Configure API key in header instead of query parameter.

6.2 Basic Authentication

Reference:

<https://learn.microsoft.com/en-us/connectors/custom-connectors/define-security#basic-authentication>

Hands-On:

Configure username/password authentication in connector.

6.3 OAuth 2.0 Configuration

Reference:

<https://learn.microsoft.com/en-us/connectors/custom-connectors/define-oauth-settings>

Hands-On:

Review OAuth fields and understand redirect URL.